

TDRP-TW definition

Considering the commercial interest and attention on both trucks and drones carried by trucks for last-mile parcel deliveries, we introduce the TDRP-TW. Solving the TDRP-TW necessitates finding the cost-minimizing routes. Three types of routes are given. The first and second types are pure drone and truck routes, in which a drone or truck route originates and terminates at the depot. The third type is combination routes. A combination route consists of one main tour and one or several sub-tours.

Each TDRP-TW network includes a number of customers and one depot housing a fleet of truck–drone combinations. The network is denoted by an undirected complete graph in which the nodes represent the depot and customers, and the arcs represent the possible moves of trucks and drones. The depot is the only parcel source. The truck–drone combination includes one truck carrying several drones. All customer demand should be satisfied. Customer demand cannot be split. The customer time windows are known in advance. Customers are classified into two types: truck customer (TC) or drone customer (UC). Each TC should be serviced by one truck, and each UC should be serviced by one drone. The number of drones carried by each truck is assumed to be the same. A vehicle in use may be a single truck or drone, or a truck–drone combination. Each truck–drone combination works in a “paired” modality. A drone can carry several parcels to serve more than one customer. It is assumed that there are enough trucks and drones. Along its route, a truck can dispatch and retrieve the carried drones only at a TC, which is referred to as the satellite. A drone can be retrieved by its truck at a satellite, or can return directly to the depot.

We choose the test instances from the Solomon (1987) VRPTW benchmark problems with 100 customers and convert them to TDRP-TW instances.

Let nc denote the number of customers included in the TDRP-TW network. The nc customers are classified into TCs and UCs. Parameter p denotes the percentage of UCs in the nc customers, and $p = 25\%$, 50% or 75% . All customer time windows are scale down by the same constant.

The demand of a TC or a UC is a random number in the range of $(0, 50 \text{ kg})$ or $(0, 2.5 \text{ kg})$, respectively. The service times at a TC and a UC are taken as 15 min and 5 min, respectively. The drone and truck capacity are 5 kg and 1000 kg, respectively. The maximum working time of each truck is 10 h, and the maximum flying time of one departure of each drone is 0.67 h. The average velocity of each truck and that of each drone are 60 km/h and 65 km/h, respectively. The variable cost of each truck is 0.8 Chinese Yuan/km, and the operating cost per departure of each drone is 15 Chinese Yuan. The penalty of a truck waiting for a unit of time at customer locations is estimated as 1.

When $nc < 100$, the converted instances are specified as small- and medium-scale instances. Each of the small- and medium-scale instances is denoted by $C“nc”-25$, $C“nc”-50$, or $C“nc”-75$; and $C“nc”-25$, $C“nc”-50$, or $C“nc”-75$ has the percentage of UCs 25%, 50%, or 75%, respectively. When $nc = 100$, the converted instances are specified as large-scale instances. Each of the large-scale instances is denoted by “name of VRPTW benchmark instance”-75, which has the percentage of UCs 75%.